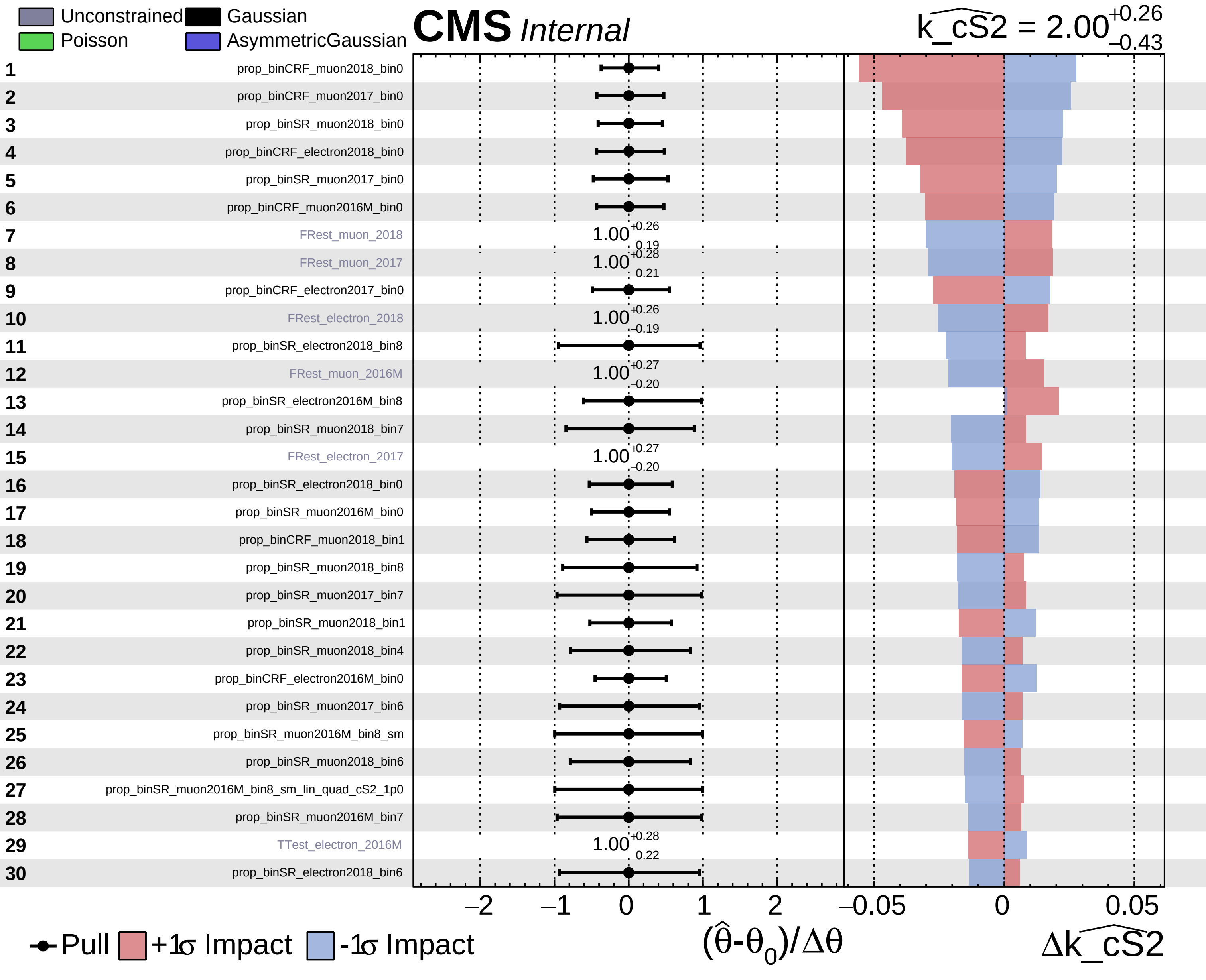
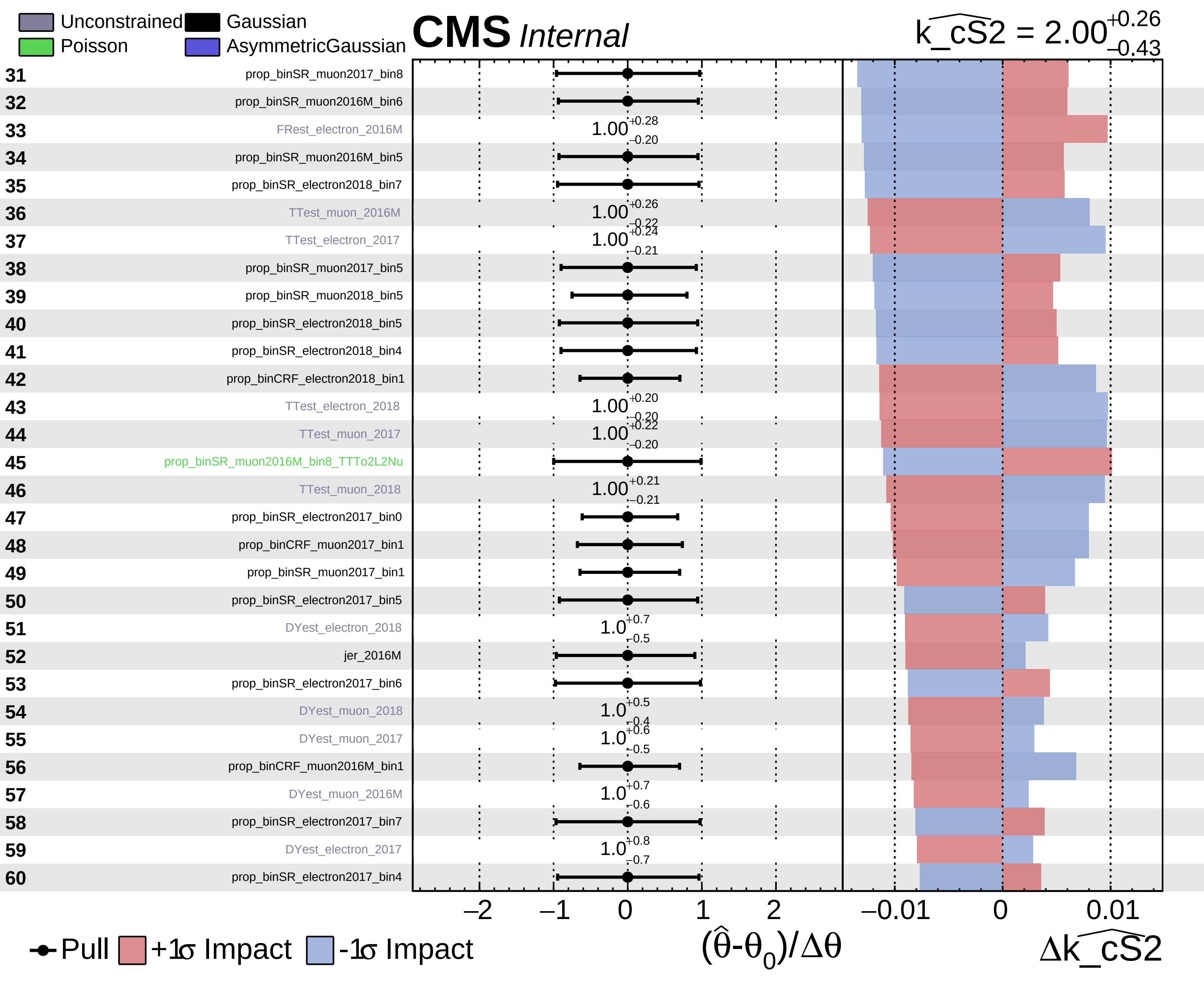
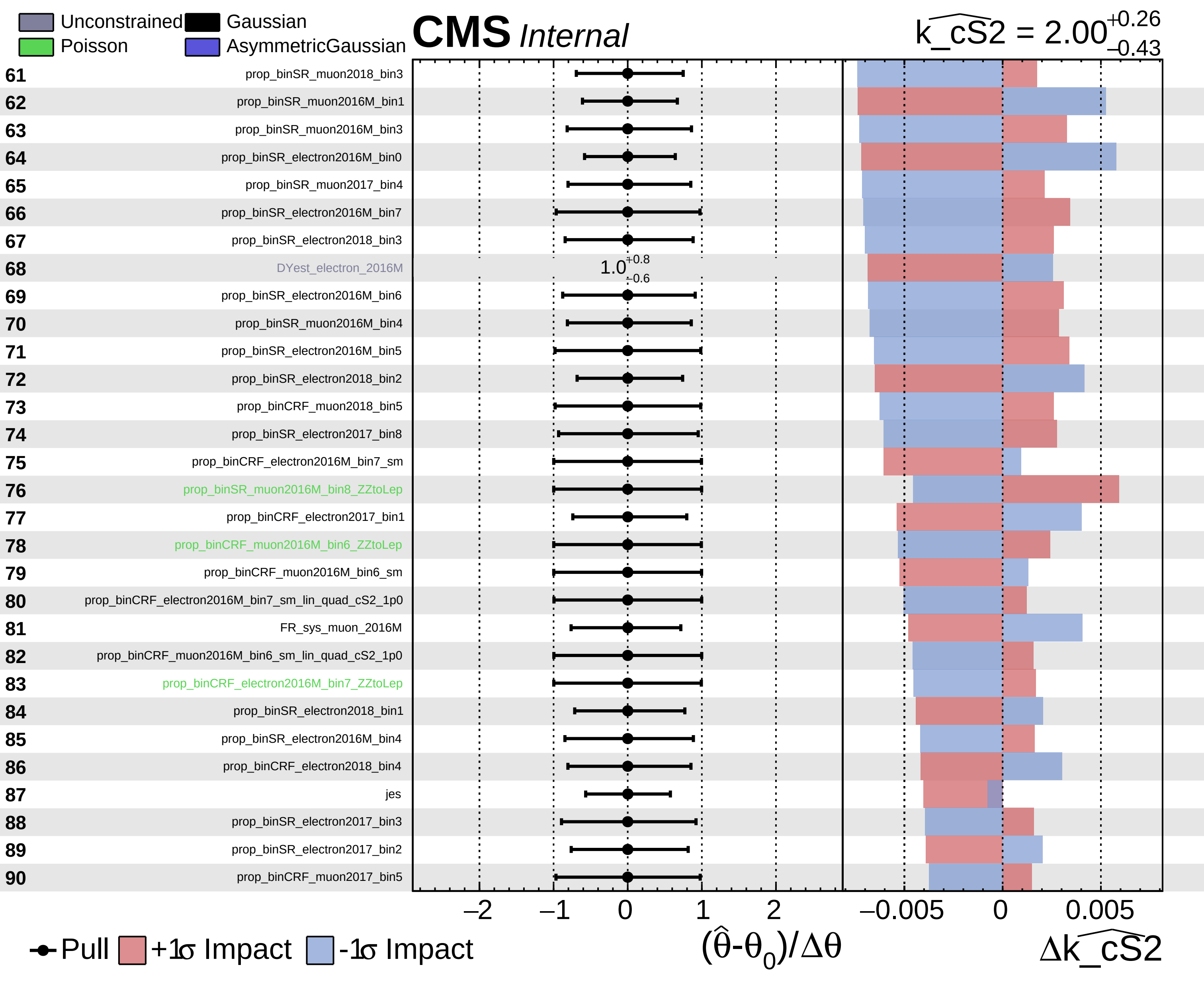
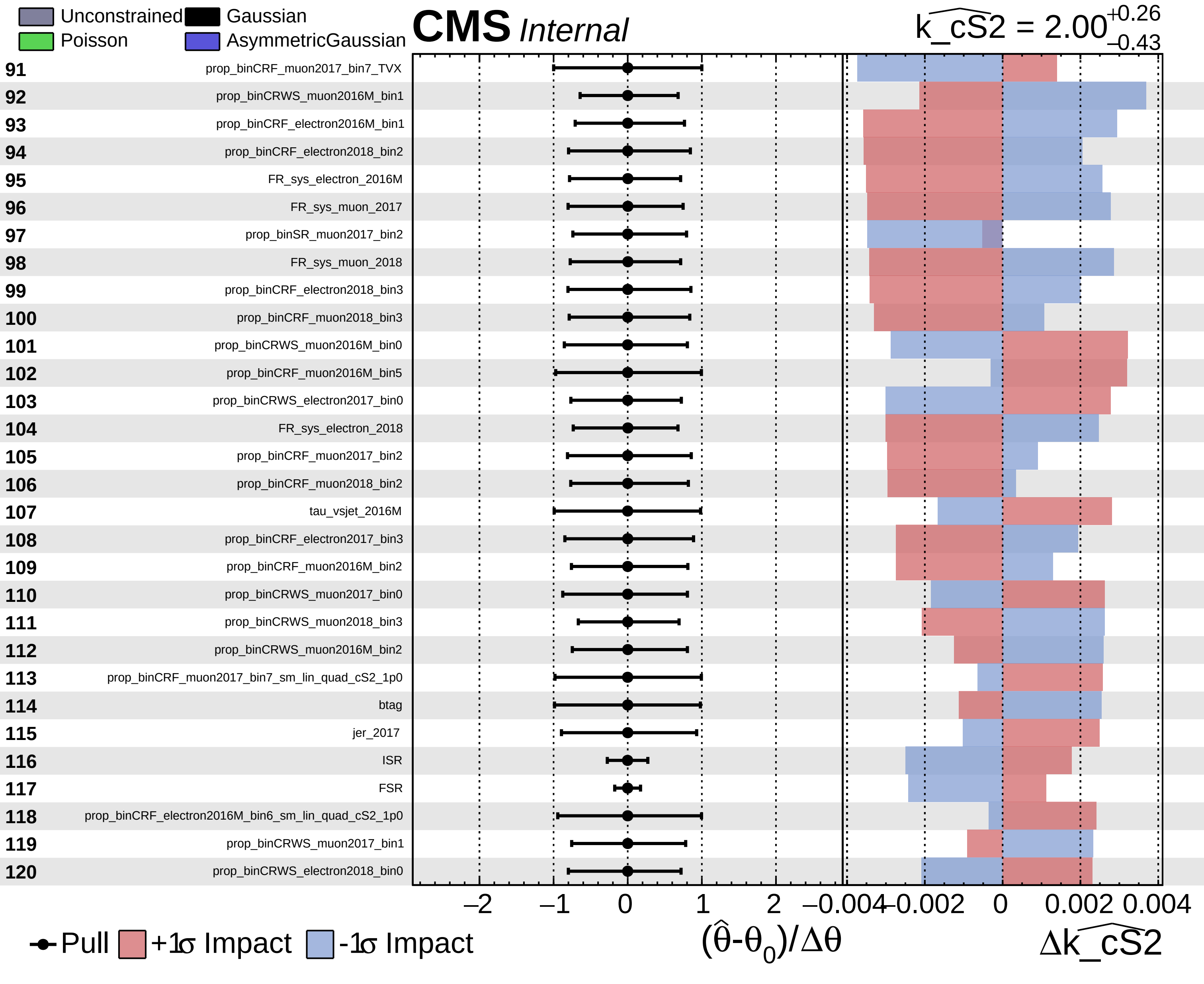

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 2.00 +0.26 −0.43
1 prop_binCRF_muon2018_bin0
2 prop_binCRF_muon2017_bin0
3 prop_binSR_muon2018_bin0
4 prop_binCRF_electron2018_bin0
5 prop_binSR_muon2017_bin0
6 prop_binCRF_muon2016M_bin0
7 FRest_muon_2018 1.00 +0.26 −0.19
8 FRest_muon_2017 1.00 +0.28 −0.21
9 prop_binCRF_electron2017_bin0
10 FRest_electron_2018 1.00 +0.26 −0.19
11 prop_binSR_electron2018_bin8
12 FRest_muon_2016M 1.00 +0.27 −0.20
13 prop_binSR_electron2016M_bin8
14 prop_binSR_muon2018_bin7
15 FRest_electron_2017 1.00 +0.27 −0.20
16 prop_binSR_electron2018_bin0
17 prop_binSR_muon2016M_bin0
18 prop_binCRF_muon2018_bin1
19 prop_binSR_muon2018_bin8
20 prop_binSR_muon2017_bin7
21 prop_binSR_muon2018_bin1
22 prop_binSR_muon2018_bin4
23 prop_binCRF_electron2016M_bin0
24 prop_binSR_muon2017_bin6
25 prop_binSR_muon2016M_bin8_sm
26 prop_binSR_muon2018_bin6
27 prop_binSR_muon2016M_bin8_sm_lin_quad_cS2_1p0
28 prop_binSR_muon2016M_bin7
29 TTest_electron_2016M 1.00 +0.28 −0.22
30 prop_binSR_electron2018_bin6
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2

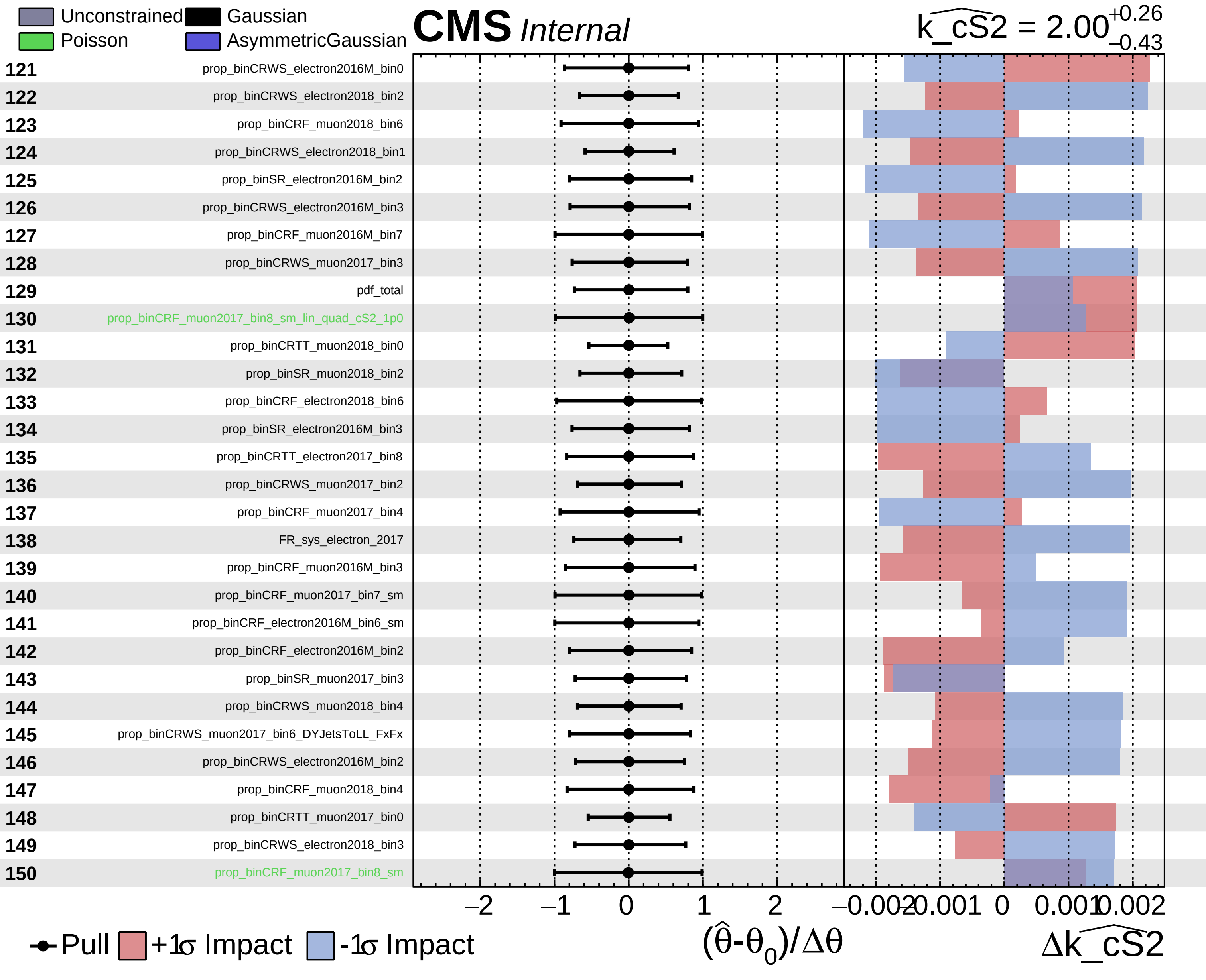


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 2.00 +0.26 −0.43
31 prop_binSR_muon2017_bin8
32 prop_binSR_muon2016M_bin6
33 FRest_electron_2016M 1.00 +0.28 −0.20
34 prop_binSR_muon2016M_bin5
35 prop_binSR_electron2018_bin7
36 TTest_muon_2016M 1.00 +0.26 −0.22
37 TTest_electron_2017 1.00 +0.24 −0.21
38 prop_binSR_muon2017_bin5
39 prop_binSR_muon2018_bin5
40 prop_binSR_electron2018_bin5
41 prop_binSR_electron2018_bin4
42 prop_binCRF_electron2018_bin1
43 TTest_electron_2018 1.00 +0.20 −0.20
44 TTest_muon_2017 1.00 +0.22 −0.20
45 prop_binSR_muon2016M_bin8_TTTo2L2Nu
46 TTest_muon_2018 1.00 +0.21 −0.21
47 prop_binSR_electron2017_bin0
48 prop_binCRF_muon2017_bin1
49 prop_binSR_muon2017_bin1
50 prop_binSR_electron2017_bin5
51 DYest_electron_2018 1.0 +0.7 −0.5
52 jer_2016M
53 prop_binSR_electron2017_bin6
54 DYest_muon_2018 1.0 +0.5 −0.4
55 DYest_muon_2017 1.0 +0.6 −0.5
56 prop_binCRF_muon2016M_bin1
57 DYest_muon_2016M 1.0 +0.7 −0.6
58 prop_binSR_electron2017_bin7
59 DYest_electron_2017 1.0 +0.8 −0.7
60 prop_binSR_electron2017_bin4
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2

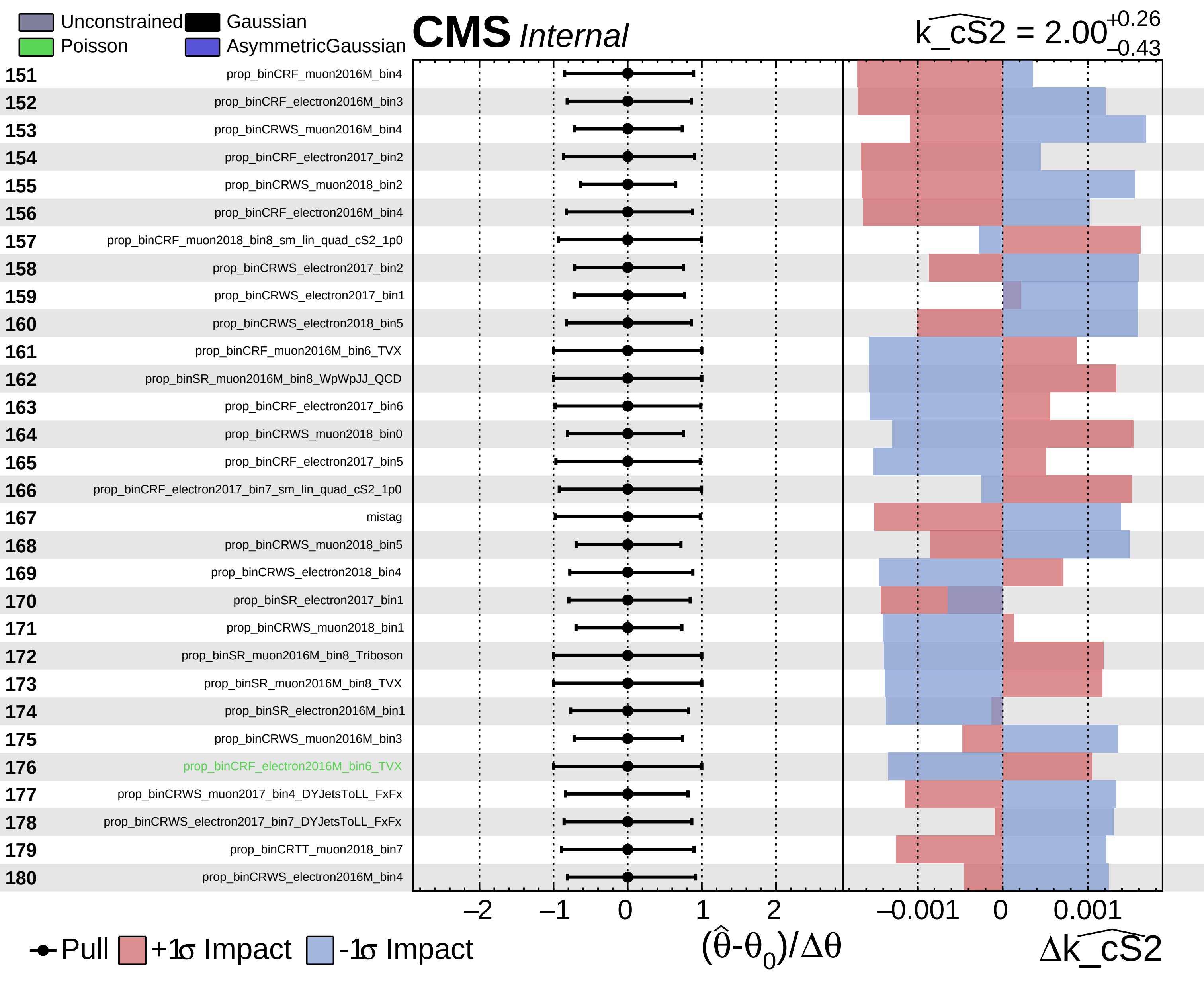

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 2.00 +0.26 −0.43
61 prop_binSR_muon2018_bin3
62 prop_binSR_muon2016M_bin1
63 prop_binSR_muon2016M_bin3
64 prop_binSR_electron2016M_bin0
65 prop_binSR_muon2017_bin4
66 prop_binSR_electron2016M_bin7
67 prop_binSR_electron2018_bin3
68 DYest_electron_2016M
1.0 +0.8 −0.6
69 prop_binSR_electron2016M_bin6
70 prop_binSR_muon2016M_bin4
71 prop_binSR_electron2016M_bin5
72 prop_binSR_electron2018_bin2
73 prop_binCRF_muon2018_bin5
74 prop_binSR_electron2017_bin8
75 prop_binCRF_electron2016M_bin7_sm
76 prop_binSR_muon2016M_bin8_ZZtoLep
77 prop_binCRF_electron2017_bin1
78 prop_binCRF_muon2016M_bin6_ZZtoLep
79 prop_binCRF_muon2016M_bin6_sm
80 prop_binCRF_electron2016M_bin7_sm_lin_quad_cS2_1p0
81 FR_sys_muon_2016M
82 prop_binCRF_muon2016M_bin6_sm_lin_quad_cS2_1p0
83 prop_binCRF_electron2016M_bin7_ZZtoLep
84 prop_binSR_electron2018_bin1
85 prop_binSR_electron2016M_bin4
86 prop_binCRF_electron2018_bin4
87 jes
88 prop_binSR_electron2017_bin3
89 prop_binSR_electron2017_bin2
90 prop_binCRF_muon2017_bin5
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2



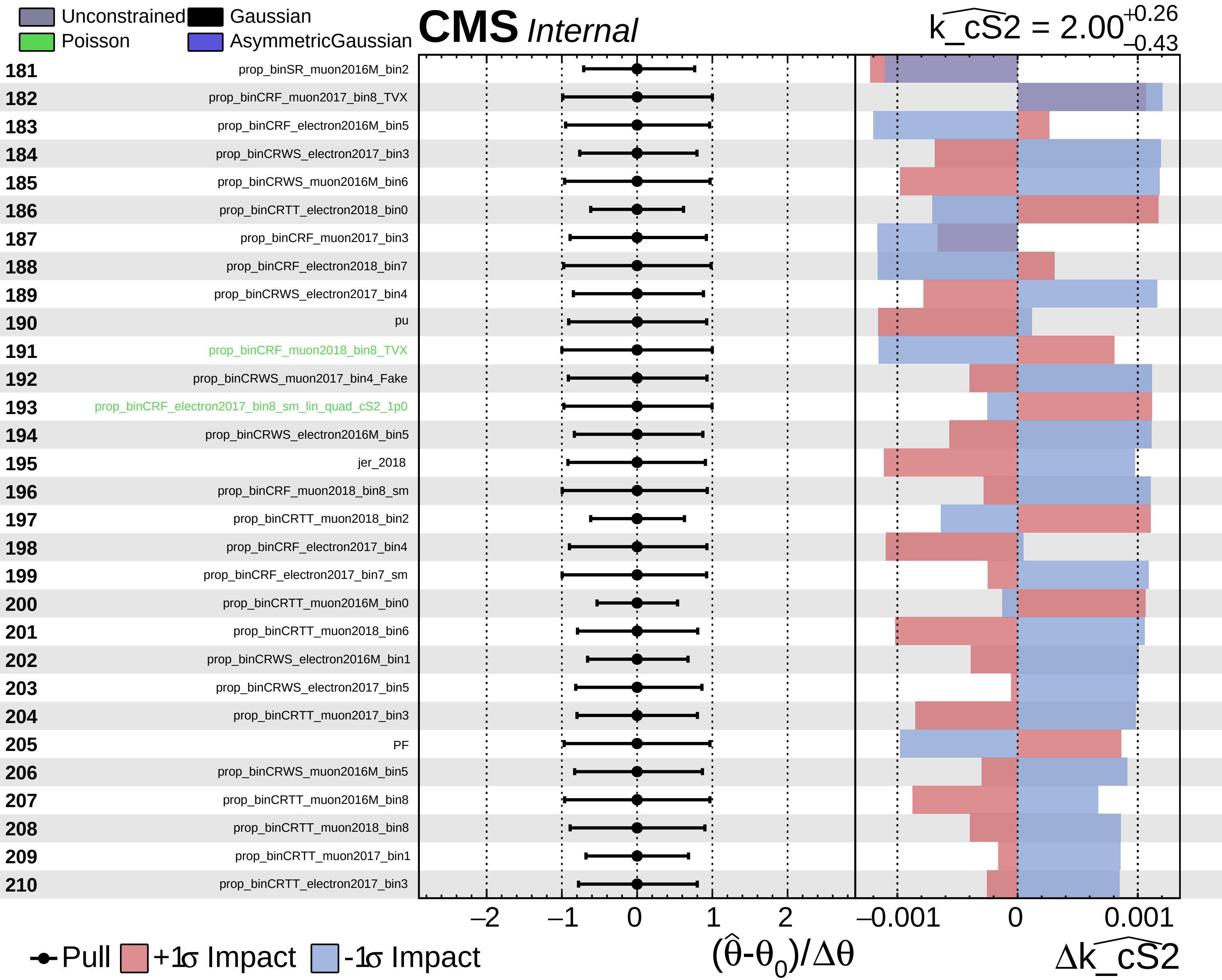
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 2.00^{+0.26}_{-0.43}$
91 prop_binCRF_muon2017_bin7_TVX
92 prop_binCRWS_muon2016M_bin1
93 prop_binCRF_electron2016M_bin1
94 prop_binCRF_electron2018_bin2
95 FR_sys_electron_2016M
96 FR_sys_muon_2017
97 prop_binSR_muon2017_bin2
98 FR_sys_muon_2018
99 prop_binCRF_electron2018_bin3
100 prop_binCRF_muon2018_bin3
101 prop_binCRWS_muon2016M_bin0
102 prop_binCRF_muon2016M_bin5
103 prop_binCRWS_electron2017_bin0
104 FR_sys_electron_2018
105 prop_binCRF_muon2017_bin2
106 prop_binCRF_muon2018_bin2
107 tau_vsjet_2016M
108 prop_binCRF_electron2017_bin3
109 prop_binCRF_muon2016M_bin2
110 prop_binCRWS_muon2017_bin0
111 prop_binCRWS_muon2018_bin3
112 prop_binCRWS_muon2016M_bin2
113 prop_binCRF_muon2017_bin7_sm_lin_quad_cS2_1p0
114 btag
115 jer_2017
116 ISR
117 FSR
118 prop_binCRF_electron2016M_bin6_sm_lin_quad_cS2_1p0
119 prop_binCRWS_muon2017_bin1
120 prop_binCRWS_electron2018_bin0
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



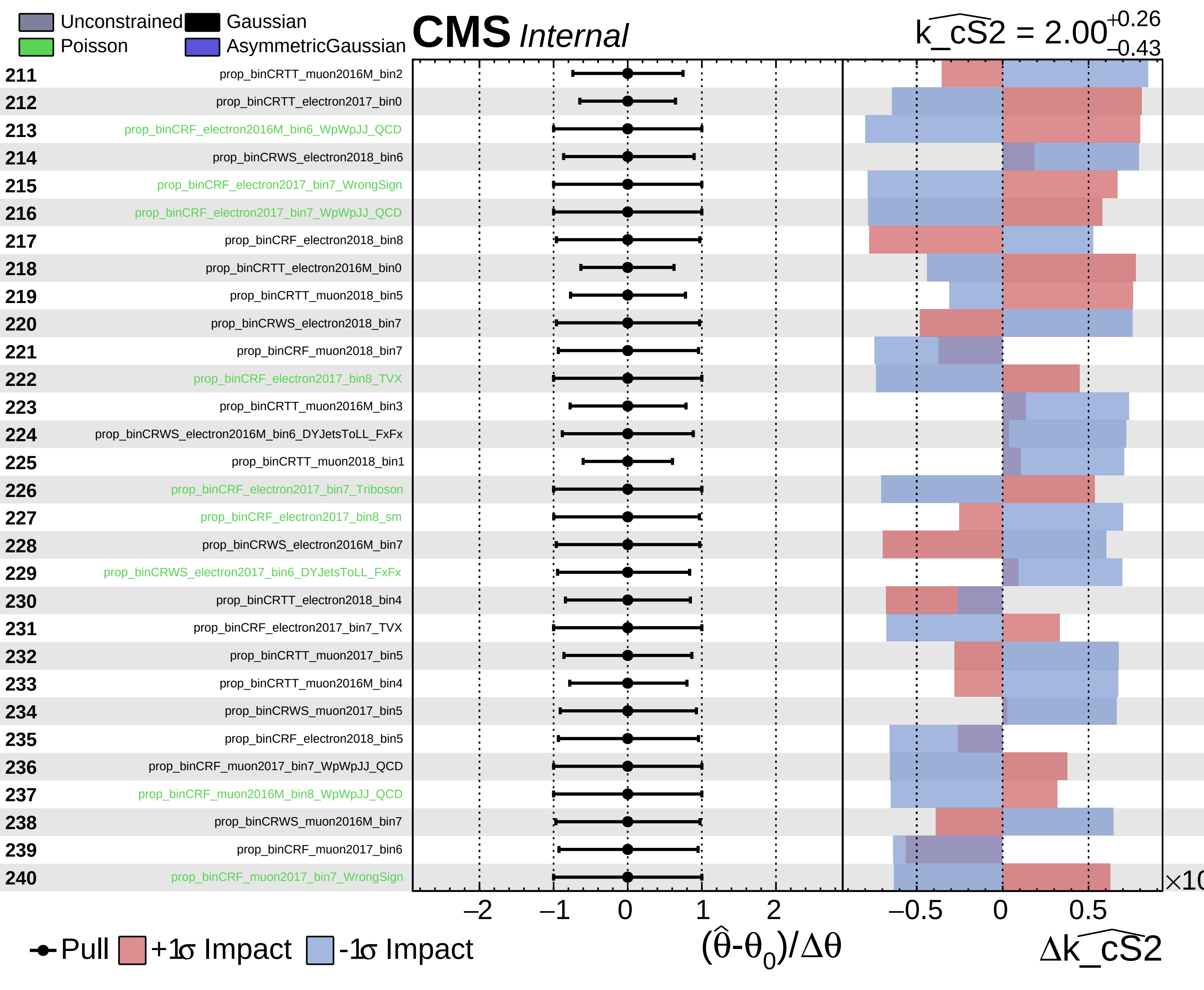
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 2.00^{+0.26}_{-0.43}$
121 prop_binCRWS_electron2016M_bin0
122 prop_binCRWS_electron2018_bin2
123 prop_binCRF_muon2018_bin6
124 prop_binCRWS_electron2018_bin1
125 prop_binSR_electron2016M_bin2
126 prop_binCRWS_electron2016M_bin3
127 prop_binCRF_muon2016M_bin7
128 prop_binCRWS_muon2017_bin3
129 pdf_total
130 prop_binCRF_muon2017_bin8_sm_lin_quad_cS2_1p0
131 prop_binCRTT_muon2018_bin0
132 prop_binSR_muon2018_bin2
133 prop_binCRF_electron2018_bin6
134 prop_binSR_electron2016M_bin3
135 prop_binCRTT_electron2017_bin8
136 prop_binCRWS_muon2017_bin2
137 prop_binCRF_muon2017_bin4
138 FR_sys_electron_2017
139 prop_binCRF_muon2016M_bin3
140 prop_binCRF_muon2017_bin7_sm
141 prop_binCRF_electron2016M_bin6_sm
142 prop_binCRF_electron2016M_bin2
143 prop_binSR_muon2017_bin3
144 prop_binCRWS_muon2018_bin4
145 prop_binCRWS_muon2017_bin6_DYJetsToLL_FxFx
146 prop_binCRWS_electron2016M_bin2
147 prop_binCRF_muon2018_bin4
148 prop_binCRTT_muon2017_bin0
149 prop_binCRWS_electron2018_bin3
150 prop_binCRF_muon2017_bin8_sm
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



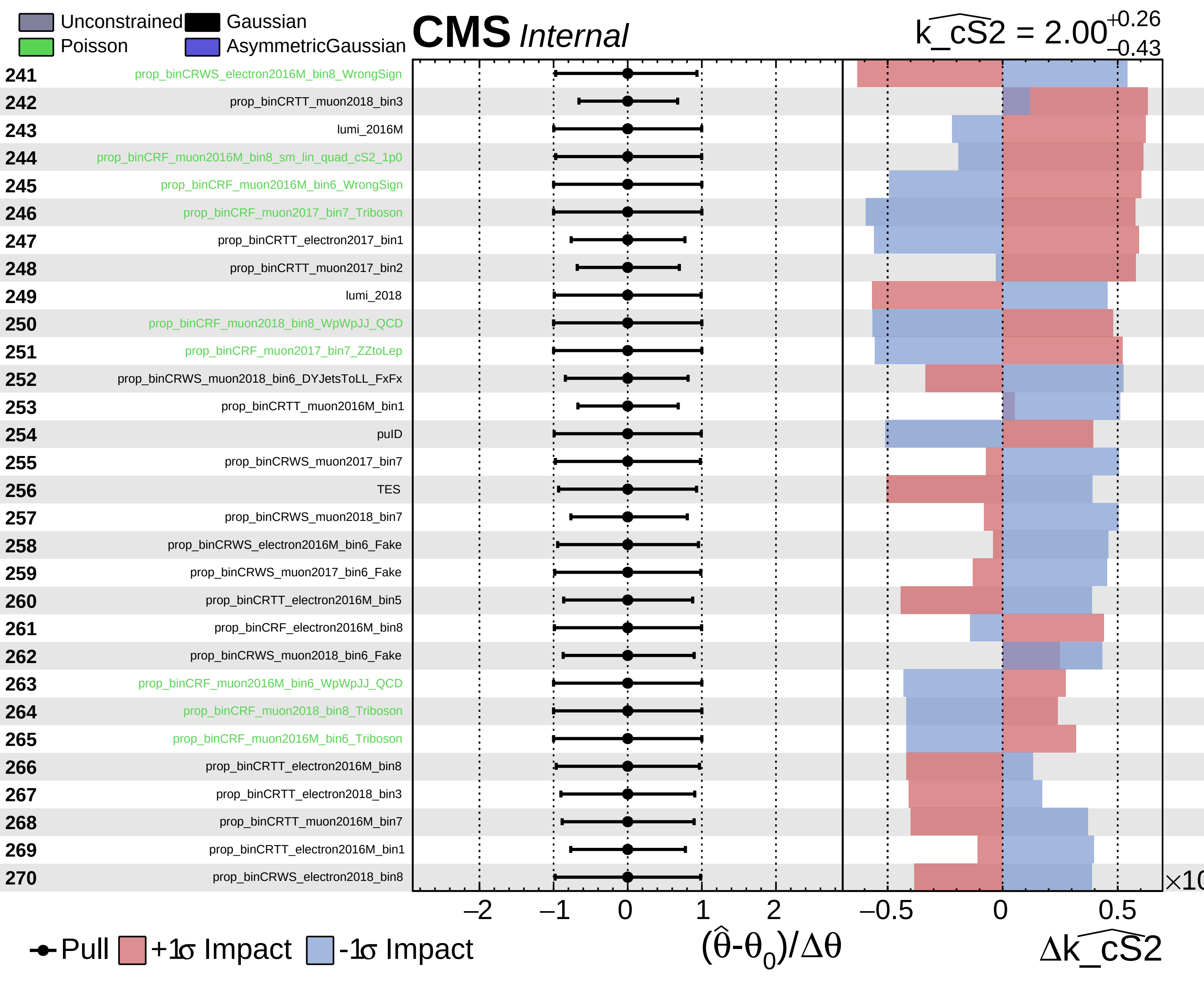
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 2.00^{+0.26}_{-0.43}$
151 prop_binCRF_muon2016M_bin4
152 prop_binCRF_electron2016M_bin3
153 prop_binCRWS_muon2016M_bin4
154 prop_binCRF_electron2017_bin2
155 prop_binCRWS_muon2018_bin2
156 prop_binCRF_electron2016M_bin4
157 prop_binCRF_muon2018_bin8_sm_lin_quad_cS2_1p0
158 prop_binCRWS_electron2017_bin2
159 prop_binCRWS_electron2017_bin1
160 prop_binCRWS_electron2018_bin5
161 prop_binCRF_muon2016M_bin6_TVX
162 prop_binSR_muon2016M_bin8_WpWpJJ_QCD
163 prop_binCRF_electron2017_bin6
164 prop_binCRWS_muon2018_bin0
165 prop_binCRF_electron2017_bin5
166 prop_binCRF_electron2017_bin7_sm_lin_quad_cS2_1p0
167 mistag
168 prop_binCRWS_muon2018_bin5
169 prop_binCRWS_electron2018_bin4
170 prop_binSR_electron2017_bin1
171 prop_binCRWS_muon2018_bin1
172 prop_binSR_muon2016M_bin8_Triboson
173 prop_binSR_muon2016M_bin8_TVX
174 prop_binSR_electron2016M_bin1
175 prop_binCRWS_muon2016M_bin3
176 prop_binCRF_electron2016M_bin6_TVX
177 prop_binCRWS_muon2017_bin4_DYJetsToLL_FxFx
178 prop_binCRWS_electron2017_bin7_DYJetsToLL_FxFx
179 prop_binCRTT_muon2018_bin7
180 prop_binCRWS_electron2016M_bin4
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



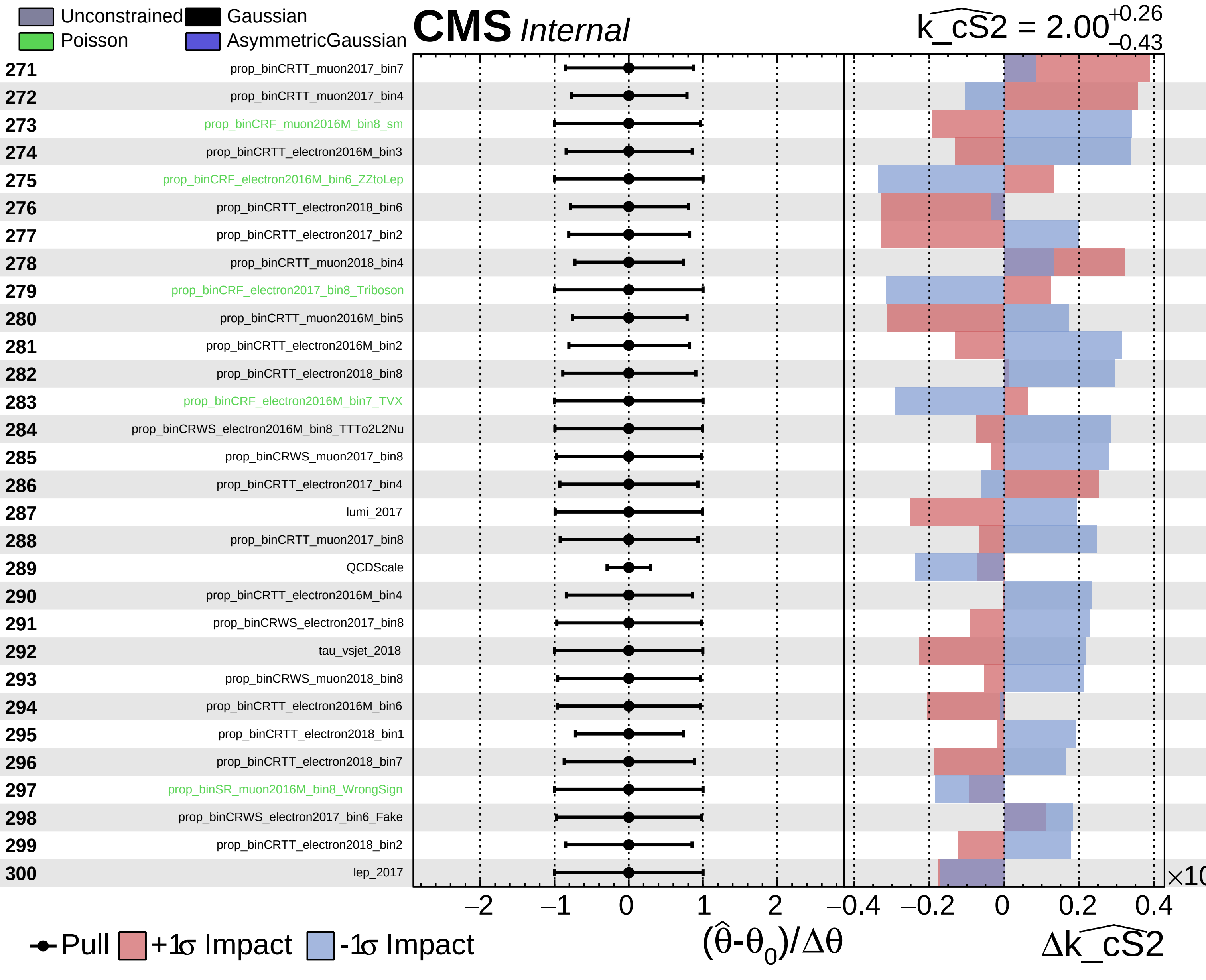
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 2.00^{+0.26}_{-0.43}$
181 prop_binSR_muon2016M_bin2
182 prop_binCRF_muon2017_bin8_TVX
183 prop_binCRF_electron2016M_bin5
184 prop_binCRWS_electron2017_bin3
185 prop_binCRWS_muon2016M_bin6
186 prop_binCRTT_electron2018_bin0
187 prop_binCRF_muon2017_bin3
188 prop_binCRF_electron2018_bin7
189 prop_binCRWS_electron2017_bin4
190 pu
191 prop_binCRF_muon2018_bin8_TVX
192 prop_binCRWS_muon2017_bin4_Fake
193 prop_binCRF_electron2017_bin8_sm_lin_quad_cS2_1p0
194 prop_binCRWS_electron2016M_bin5
195 jer_2018
196 prop_binCRF_muon2018_bin8_sm
197 prop_binCRTT_muon2018_bin2
198 prop_binCRF_electron2017_bin4
199 prop_binCRF_electron2017_bin7_sm
200 prop_binCRTT_muon2016M_bin0
201 prop_binCRTT_muon2018_bin6
202 prop_binCRWS_electron2016M_bin1
203 prop_binCRWS_electron2017_bin5
204 prop_binCRTT_muon2017_bin3
205 PF
206 prop_binCRWS_muon2016M_bin5
207 prop_binCRTT_muon2016M_bin8
208 prop_binCRTT_muon2018_bin8
209 prop_binCRTT_muon2017_bin1
210 prop_binCRTT_electron2017_bin3
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

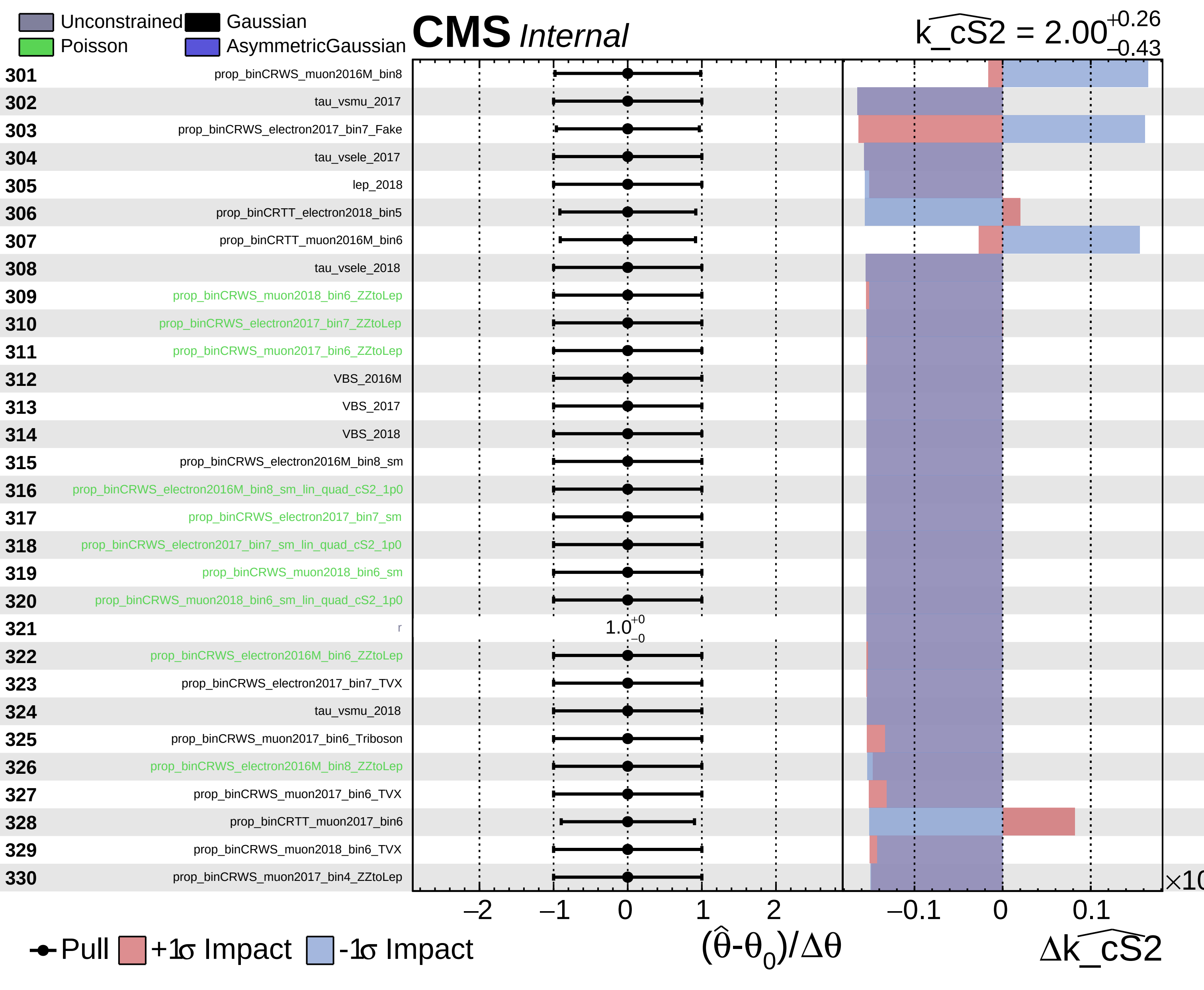


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 2.00 +0.26 −0.43
211 prop_binCRTT_muon2016M_bin2
212 prop_binCRTT_electron2017_bin0
213 prop_binCRF_electron2016M_bin6_WpWpJJ_QCD
214 prop_binCRWS_electron2018_bin6
215 prop_binCRF_electron2017_bin7_WrongSign
216 prop_binCRF_electron2017_bin7_WpWpJJ_QCD
217 prop_binCRF_electron2018_bin8
218 prop_binCRTT_electron2016M_bin0
219 prop_binCRTT_muon2018_bin5
220 prop_binCRWS_electron2018_bin7
221 prop_binCRF_muon2018_bin7
222 prop_binCRF_electron2017_bin8_TVX
223 prop_binCRTT_muon2016M_bin3
224 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
225 prop_binCRTT_muon2018_bin1
226 prop_binCRF_electron2017_bin7_Triboson
227 prop_binCRF_electron2017_bin8_sm
228 prop_binCRWS_electron2016M_bin7
229 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
230 prop_binCRTT_electron2018_bin4
231 prop_binCRF_electron2017_bin7_TVX
232 prop_binCRTT_muon2017_bin5
233 prop_binCRTT_muon2016M_bin4
234 prop_binCRWS_muon2017_bin5
235 prop_binCRF_electron2018_bin5
236 prop_binCRF_muon2017_bin7_WpWpJJ_QCD
237 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
238 prop_binCRWS_muon2016M_bin7
239 prop_binCRF_muon2017_bin6
240 prop_binCRF_muon2017_bin7_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

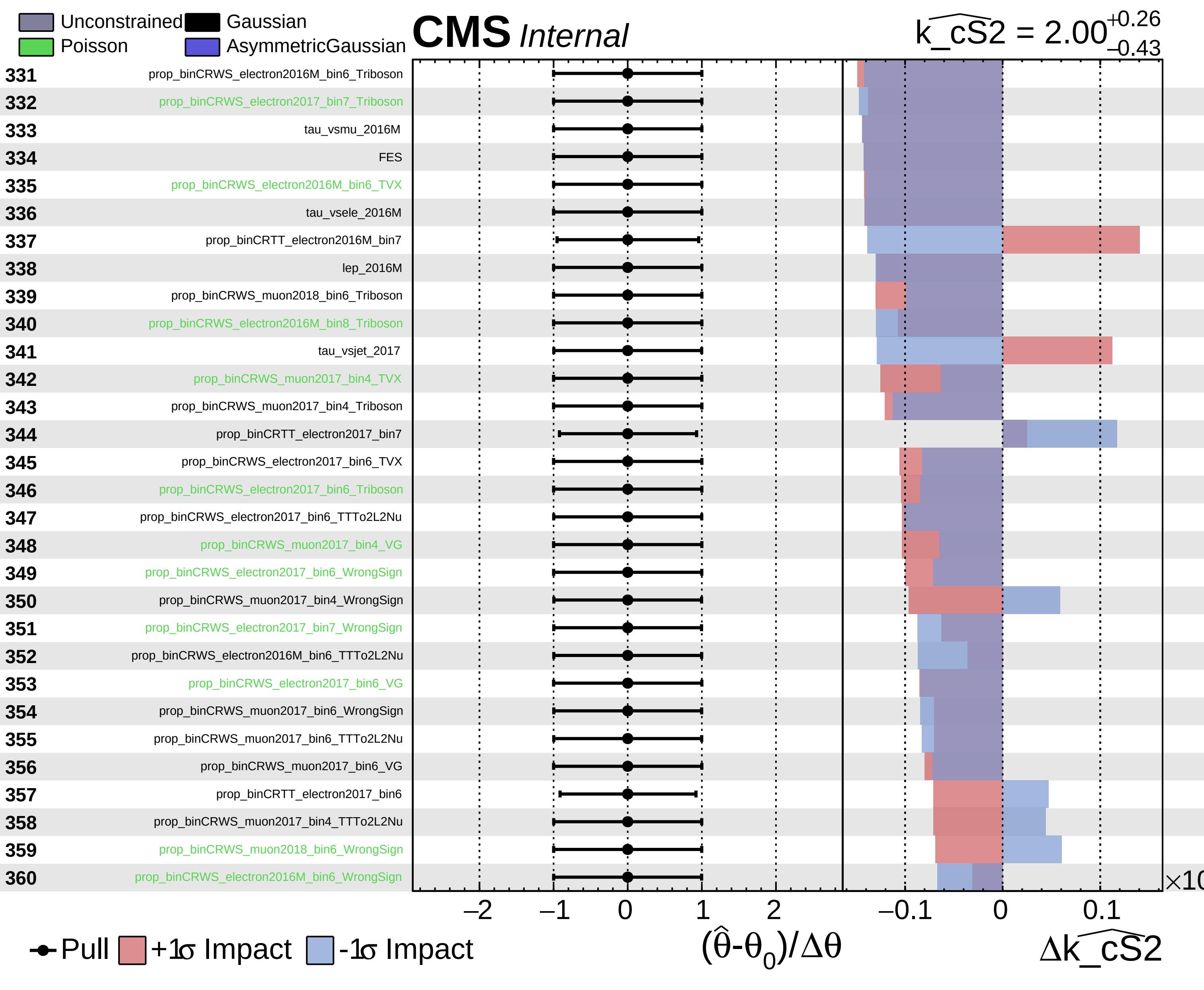


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 2.00^{+0.26}_{-0.43}$
241 prop_binCRWS_electron2016M_bin8_WrongSign
242 prop_binCRTT_muon2018_bin3
243 lumi_2016M
244 prop_binCRF_muon2016M_bin8_sm_lin_quad_cS2_1p0
245 prop_binCRF_muon2016M_bin6_WrongSign
246 prop_binCRF_muon2017_bin7_Triboson
247 prop_binCRTT_electron2017_bin1
248 prop_binCRTT_muon2017_bin2
249 lumi_2018
250 prop_binCRF_muon2018_bin8_WpWpJJ_QCD
251 prop_binCRF_muon2017_bin7_ZZtoLep
252 prop_binCRWS_muon2018_bin6_DYJetsToLL_FxFx
253 prop_binCRTT_muon2016M_bin1
254 puID
255 prop_binCRWS_muon2017_bin7
256 TES
257 prop_binCRWS_muon2018_bin7
258 prop_binCRWS_electron2016M_bin6_Fake
259 prop_binCRWS_muon2017_bin6_Fake
260 prop_binCRTT_electron2016M_bin5
261 prop_binCRF_electron2016M_bin8
262 prop_binCRWS_muon2018_bin6_Fake
263 prop_binCRF_muon2016M_bin6_WpWpJJ_QCD
264 prop_binCRF_muon2018_bin8_Triboson
265 prop_binCRF_muon2016M_bin6_Triboson
266 prop_binCRTT_electron2016M_bin8
267 prop_binCRTT_electron2018_bin3
268 prop_binCRTT_muon2016M_bin7
269 prop_binCRTT_electron2016M_bin1
270 prop_binCRWS_electron2018_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$

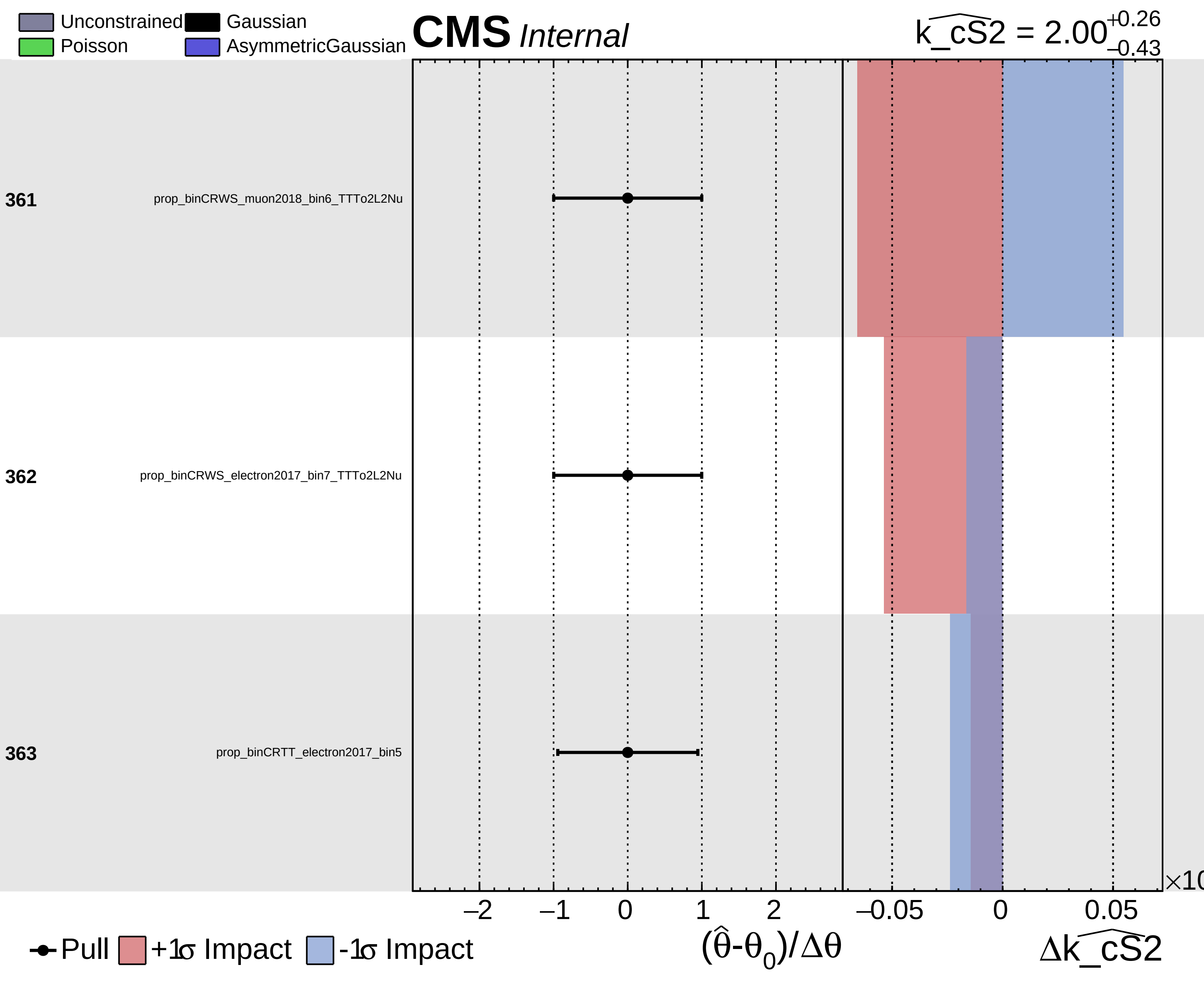

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 2.00 +0.26 −0.43
271 prop_binCRTT_muon2017_bin7
272 prop_binCRTT_muon2017_bin4
273 prop_binCRF_muon2016M_bin8_sm
274 prop_binCRTT_electron2016M_bin3
275 prop_binCRF_electron2016M_bin6_ZZtoLep
276 prop_binCRTT_electron2018_bin6
277 prop_binCRTT_electron2017_bin2
278 prop_binCRTT_muon2018_bin4
279 prop_binCRF_electron2017_bin8_Triboson
280 prop_binCRTT_muon2016M_bin5
281 prop_binCRTT_electron2016M_bin2
282 prop_binCRTT_electron2018_bin8
283 prop_binCRF_electron2016M_bin7_TVX
284 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
285 prop_binCRWS_muon2017_bin8
286 prop_binCRTT_electron2017_bin4
287 lumi_2017
288 prop_binCRTT_muon2017_bin8
289 QCDScale
290 prop_binCRTT_electron2016M_bin4
291 prop_binCRWS_electron2017_bin8
292 tau_vsjet_2018
293 prop_binCRWS_muon2018_bin8
294 prop_binCRTT_electron2016M_bin6
295 prop_binCRTT_electron2018_bin1
296 prop_binCRTT_electron2018_bin7
297 prop_binSR_muon2016M_bin8_WrongSign
298 prop_binCRWS_electron2017_bin6_Fake
299 prop_binCRTT_electron2018_bin2
300 lep_2017
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 2.00 +0.26 −0.43
301 prop_binCRWS_muon2016M_bin8
302 tau_vsmu_2017
303 prop_binCRWS_electron2017_bin7_Fake
304 tau_vsele_2017
305 lep_2018
306 prop_binCRTT_electron2018_bin5
307 prop_binCRTT_muon2016M_bin6
308 tau_vsele_2018
309 prop_binCRWS_muon2018_bin6_ZZtoLep
310 prop_binCRWS_electron2017_bin7_ZZtoLep
311 prop_binCRWS_muon2017_bin6_ZZtoLep
312 VBS_2016M
313 VBS_2017
314 VBS_2018
315 prop_binCRWS_electron2016M_bin8_sm
316 prop_binCRWS_electron2016M_bin8_sm_lin_quad_cS2_1p0
317 prop_binCRWS_electron2017_bin7_sm
318 prop_binCRWS_electron2017_bin7_sm_lin_quad_cS2_1p0
319 prop_binCRWS_muon2018_bin6_sm
320 prop_binCRWS_muon2018_bin6_sm_lin_quad_cS2_1p0
321 r
1.0 +0 −0
322 prop_binCRWS_electron2016M_bin6_ZZtoLep
323 prop_binCRWS_electron2017_bin7_TVX
324 tau_vsmu_2018
325 prop_binCRWS_muon2017_bin6_Triboson
326 prop_binCRWS_electron2016M_bin8_ZZtoLep
327 prop_binCRWS_muon2017_bin6_TVX
328 prop_binCRTT_muon2017_bin6
329 prop_binCRWS_muon2018_bin6_TVX
330 prop_binCRWS_muon2017_bin4_ZZtoLep
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS2} = 2.00^{+0.26}_{-0.43}$
331 prop_binCRWS_electron2016M_bin6_Triboson
332 prop_binCRWS_electron2017_bin7_Triboson
333 tau_vsmu_2016M
334 FES
335 prop_binCRWS_electron2016M_bin6_TVX
336 tau_vsele_2016M
337 prop_binCRTT_electron2016M_bin7
338 lep_2016M
339 prop_binCRWS_muon2018_bin6_Triboson
340 prop_binCRWS_electron2016M_bin8_Triboson
341 tau_vsjet_2017
342 prop_binCRWS_muon2017_bin4_TVX
343 prop_binCRWS_muon2017_bin4_Triboson
344 prop_binCRTT_electron2017_bin7
345 prop_binCRWS_electron2017_bin6_TVX
346 prop_binCRWS_electron2017_bin6_Triboson
347 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
348 prop_binCRWS_muon2017_bin4_VG
349 prop_binCRWS_electron2017_bin6_WrongSign
350 prop_binCRWS_muon2017_bin4_WrongSign
351 prop_binCRWS_electron2017_bin7_WrongSign
352 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
353 prop_binCRWS_electron2017_bin6_VG
354 prop_binCRWS_muon2017_bin6_WrongSign
355 prop_binCRWS_muon2017_bin6_TTTo2L2Nu
356 prop_binCRWS_muon2017_bin6_VG
357 prop_binCRTT_electron2017_bin6
358 prop_binCRWS_muon2017_bin4_TTTo2L2Nu
359 prop_binCRWS_muon2018_bin6_WrongSign
360 prop_binCRWS_electron2016M_bin6_WrongSign
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS2}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS2 = 2.00 +0.26 −0.43
361
prop_binCRWS_muon2018_bin6_TTTo2L2Nu
362
prop_binCRWS_electron2017_bin7_TTTo2L2Nu
363
prop_binCRTT_electron2017_bin5
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS2